



3M's M2167PW multi-touch display offers 20 simultaneous touch events at an ultra-fast 6-millisecond response time

is responsible for the delay we notice between giving the input and seeing it on screen. In a video demonstration, a Microsoft researcher showed a 10ms latency screen that changed the delay of a movable virtual object on the screen to a rubber band impression. This was followed by a 1ms delay version, in which the delay was barely perceivable. However, the company also made it clear that this is still a 'technology' and will take many years before it becomes a product.

Sun-screen, anybody?

Readability in sunlight is an important parameter for comparison of mobile device models these days and justly so. This requirement holds even for outdoor displays used for events or promotions.

To improve visibility in sunlight, some general guidelines are that a sun-readable display should be bright enough to be clearly visible under direct sunlight. Its display contrast ratio should also be high. Generally, the screen brightness is of the order of 1000 nits and the contrast ratio is at least 5:1. Plus, some technology should be used to avoid the glare.

Some companies use the combination of a standard display, an enhancement film applied on the internal screen to optimise brightness and contrast, and a polarising film applied on both sides of the touch screen to minimise

reflection. The anti-reflective coated protection glass minimises the mirror image and maximises transmittance, while the anti-glare coated protection glass scatters light directed on the surface and softens the image of direct light sources.

Pixel Qi's transfective technology is one such technology worth watching. It uses a special layer that acts like a partial mirror—part of it transmits the light while the rest of it reflects the light.

One industry expert writes: "It's like a typical

mirror layer found in most displays but with holes punched out in it. This allows for a composite mode where users can adjust the backlight and still see the LCD screen with reasonable brightness."

This transfective LCD is viewable under all lighting conditions including direct sunlight regardless of the LCD's original brightness. Also, because the LCD pixels are reflective, the user can turn down the brightness level and save on battery power consumption.

Tru-Vu uses optical bonding to improve daylight viewing on their LCDs. Optical bonding injects a transparent optical-grade epoxy resin to bond a protective glass panel directly to the face of the LCD. This eliminates the air gap between the LCD and the cover glass, drastically reducing the reflections which cause the visual wash-out typically seen in bright conditions. Optical bonding also increases the durability of the LCD monitor by bonding the LCD panel to the cover glass and the monitor's enclosure. It also avoids fogging due to moisture trapped between the layers.

Eco angles—power, heat, greenness

Users want to use their mobile devices for longer hours on a single charge. So it is critical that these devices consume less power. For larger displays, it is a matter of reducing electricity bills.

Hence companies are coming up with several innovative technologies to reduce the power consumed.

For example, this November, Japan Display showcased a reflective LCD panel with power consumption as low as 3 mW at the time of displaying a still image. The panel has a 17.8cm screen and is targeted at mobile devices that require low power consumption.

Japan Display has used a low-temperature polycrystalline silicon TFT as the driver element. It has also included a static RAM circuit on each pixel to retain image data to be written, with the aim of lowering power consumption. Another distinctive aspect of this display is that the company has achieved a natural display like paper by optimising the optical design of the scattering layer.

Materials engineering is also contributing a lot to making cooler and power-efficient displays. An organic LED display, for example, if designed properly uses only 40 per cent of the power of a comparable LCD. Nanotechnology has also given rise to better structures, materials and technologies. Cambridge University's polymer organic light-emitting diode (P-OLED) display is an example, which is extremely eco-friendly and power-efficient. P-OLEDs are basically OLEDs made with conjugated polymers.

The heat dissipated by the display is also a critical factor. If a display heats up, its basic components wear out faster and the life of the display goes down. If a lot of heat emanates from the display, it will also affect the surrounding parts of the device—this is very critical for mobile devices where the components are packed closely together. A hot device is also difficult to hold in your hand. So this is another point of innovation for display-makers.

Greenness is a universal parameter that almost all industries are focusing on. The eco-friendliness of products is a very important factor as it influences product ratings, customer decisions and the manufacturer's social responsibility in general. At GITEX 2012, Panasonic was apparently the largest exhibitor and its focus was on eco-friendly prod-